**Project Name: -** STUDENT ATTENDANCE MANAGEMENT SYSTEM

**Team Member:-**

**-Sarita**

**-Nazmunisha Sheik Abdul Rasheed**

**Introduction:-**

The purpose of the Student Attendance Management System is to automate the processing of recording, managing, and monitoring student attendance. The system replaces manual attendance registers with a digital solution that improves efficiency, accuracy, and accessibility.

**Waterfall Development Flow**

Requirements Analysis
↓
System Design
↓
Implementation (Coding)
↓
Testing
↓
Deployment (Live Cloud Hosting)
↓
Maintenance

**Phase 1: Requirements Analysis**

In this phase, all functional and non-functional requirements of the system are gathered, analysed, and documented.

- System will be built using **Python**
- Data will be stored in **MySQL**

**Functional Requirements**

- Student registration and management.
- Teacher/Admin secure login.
- Attendance marking and tracking.
- Attendance history management.
- Automated report generation.
- Search and filtering functionality.
- Attendance analytics dashboard.
- Automated facial recognition attendance using a laptop/mobile camera.
- List of absent students.
- Class-wise attendance percentage calculation.

**Non-Functional Requirements**

- Security and authentication.
- Data accuracy and reliability.

**Phase 2: System Design**

The system architecture, database structure, and user interface designs are prepared.

**Design Activities**

- Design system architecture.
- Create database tables for students and attendance records.
- Design login, attendance, and report screens.
- Define data flow and system workflow
    - Create Class diagram, UML Diagram, sequence Diagram, and ER Diagram

**Phase 3: Implementation (Coding)**

The system is developed in Python in accordance with the design specifications.

**Development Tasks**

- User authentication module.
- Student management module.
- Attendance marking module.
- Report generation module.
- Search and filtering functionality.
- Dashboard and analytics module.

**Phase 4: Testing**

- Python + MySQL connection
- Data saving correctly

- Facial recognition accuracy.
- Attendance marking functionality.
- Dashboard calculations and charts.
- User acceptance testing.

**Phase 5: Deployment**

Deploy the live system using a free cloud platform such as:

-GitHub

**Phase 6: Maintenance**

After deployment, the system is monitored and maintained.

**Maintenance Activities**

- Fix software bugs.
- Update security features.
- Improve performance.
- Add future enhancements.